



Navigating the Data Use & Privacy Tradeoff

An Information-Theoretic Approach to Privacy Regulation

Ayelet Gordon-Tapiero (Law Faculty & Dept. of Computer Science, Hebrew University); **Katrina Ligett** (Dept. of Computer Science, Hebrew University); **Kobbi Nissim** (Dept. of Computer Science, Georgetown University); **Muhammad Saad** (McCourt School, Georgetown University)

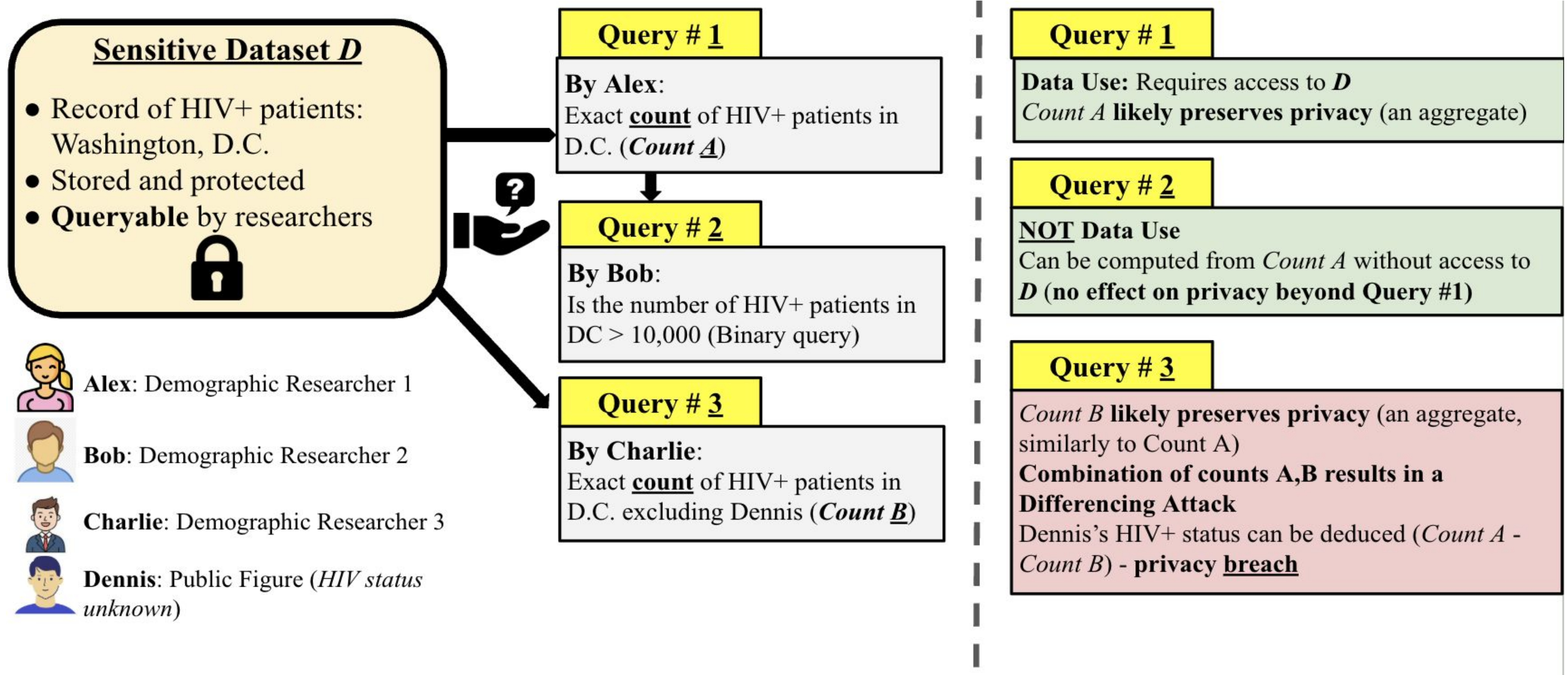
Research Motivation:

- ★ **Practical Motivation:**
Data privacy regulation should account for **disclosure risks** that **accumulate** from **multiple uses of data**, and **across parties, sources** and **timelines**.
- ★ **Theoretical Motivation:**
Understand **economic** and **information theoretic** properties of data use and privacy

Current Regulation:

- ❖ U.S. regulation recognizes that privacy harms accumulate with use of data
- ❖ **Mosaic Effect:** Individual data releases may be harmless, but could lead to privacy disclosure when viewed with other sources
- ❖ **Unit of regulation:** Individual dataset at the point of release

- ### Contributions:
1. General theory of data use and impact on privacy harms
 2. Application of information theory & privacy-related mathematical frameworks
 3. Develop a definition of **data use** and how privacy harms accumulate through examples



Recommended Regulatory Perspective:

- ❖ Coordinate data use across users, sources and timeline instead of individual releases
- ❖ Use programmable mathematical frameworks about privacy to coordinate how data is used and privacy depleted
- ❖ Regulate data use as a rival good. Privacy depletes with multiple data uses, hence, need to manage the scarcity.

Recommended Regulatory Principles:

- 📍 **Track/coordinate across different uses/users:**
Manage cumulative privacy loss through a shared and targeted privacy budget
- ⚡ **Reduction of Data Use:**
Prioritize low privacy-cost uses, regulate access and accept lower accuracy in outputs
- 📄 **Use Synthetic Data:**
Synthetic data generated with mathematical guarantees can be reused without privacy loss
- 👥 **Institutional Coordination:**
Appoint data curators to track, allocate, and enforce privacy budgets across users